# Titanic Exploratory Data Analysis & Business Insights Report

*Project 2 — Data Analyst Track*

Prepared by: Muhammad Hassan Shah
BS Software Engineering, UET Mardan
Dataset: Titanic (Kaggle Competition)

# Part 1: Data Understanding

The Titanic dataset contains passenger-level records from the RMS Titanic, which sank on 15 April 1912 after striking an iceberg during its maiden voyage. After the cleaning steps carried out in Project 1 — imputing missing **Age** values using the median age of each Passenger Class / Sex group, filling missing **Embarked** values with the most frequent port, and converting the mostly-missing **Cabin** column into a binary **HasCabin** flag — the dataset contains **891 rows and 11 columns**.

The available features span four broad categories: demographic information (Age, Sex), socio-economic indicators (Passenger Class, Fare, HasCabin), family composition (Siblings/Spouses, Parents/Children), and voyage details (Port of Embarkation), alongside the target variable **Survived** (0 = did not survive, 1 = survived). The dataset is clean, with zero missing values remaining and zero duplicate rows, and is well suited to survival-pattern analysis.

## Dataset Summary

| Metric | Value |
|---|---|
| Total Passengers (rows) | 891 |
| Total Columns | 11 |
| Missing Values (after cleaning) | 0 |
| Duplicate Rows | 0 |
| Overall Survival Rate | 38.4% |

# Part 2: Feature Engineering

Four new features were engineered from the existing columns to support deeper analysis:

### 1. Family Size

**FamilySize = SibSp + Parch + 1.** Combining siblings/spouses and parents/children into one total makes it possible to study how the overall size of a travel group affected survival, rather than two separate partial counts.

### 2. Is Alone

A binary flag (Yes / No) derived from FamilySize == 1. Traveling alone versus with family may affect how quickly a passenger reached a lifeboat, so this turns that comparison into a single simple lookup.

### 3. Age Group

Passengers were bucketed into **Child (0–12), Teenager (13–19), Adult (20–59), and Senior Citizen (60+)**. This reflects the real-world "women and children first" evacuation policy and makes life-stage survival comparisons much clearer than raw continuous age.

### 4. Fare Category

Fares were bucketed into **Low Fare (≤ £15), Medium Fare (£15–£60), and High Fare (> £60)**, chosen from the shape of the actual fare distribution. Fare is a proxy for socio-economic status and cabin location, so these bands make spending-tier comparisons straightforward.

### Engineered Feature Distributions

| Feature | Categories (count) |
|---|---|
| Age Group | Adult: 701, Teenager: 95, Child: 69, Senior Citizen: 26 |
| Fare Category | Low Fare: 458, Medium Fare: 311, High Fare: 122 |
| Is Alone | Yes: 537, No: 354 |

# Part 3: Exploratory Data Analysis — Business Questions

## Question 1: What percentage of passengers survived?

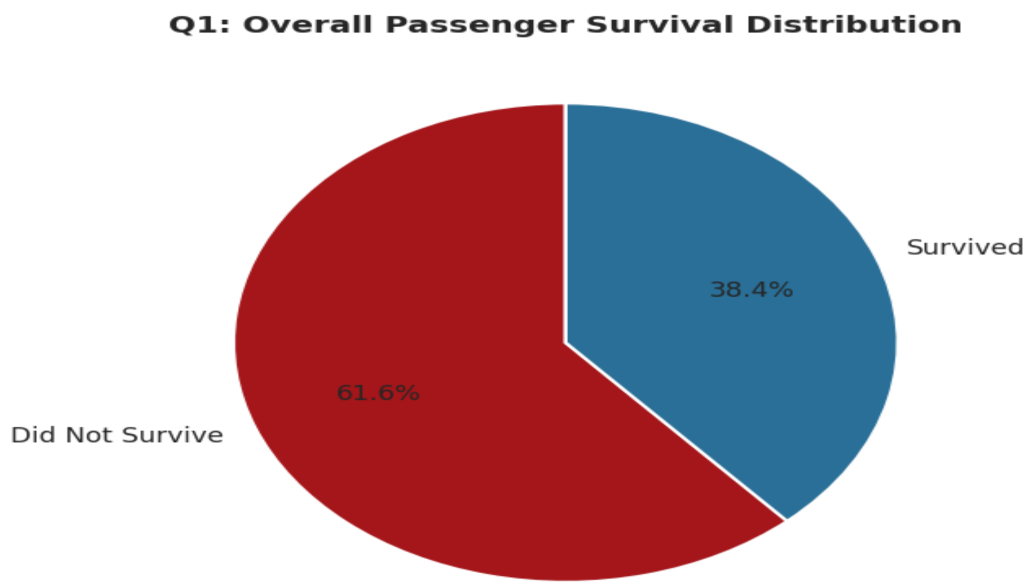


*Figure 1. Overall passenger survival distribution.*

***Observation:*** *Only 38.4% of the 891 passengers survived, while 61.6% did not. This confirms the disaster was highly fatal overall and sets the baseline for every subgroup comparison that follows.*

## Question 2: Which gender had the highest survival rate?

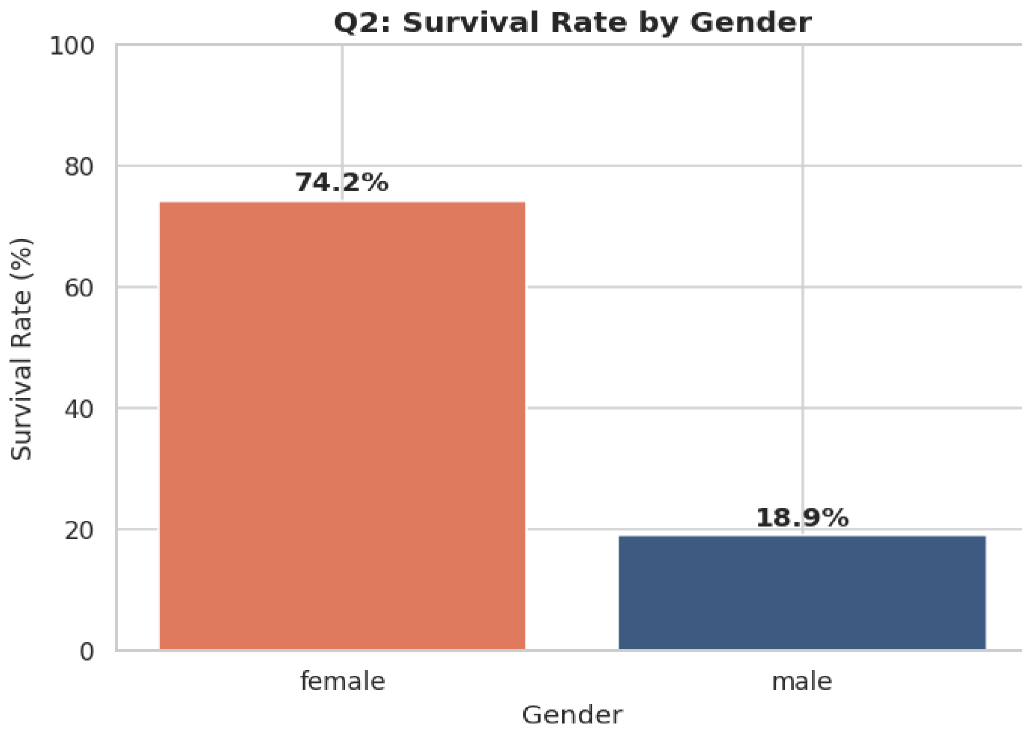


*Figure 2. Survival rate by gender.*

***Observation:*** *Women survived at 74.2%, nearly four times the male survival rate of 18.9%. This is the strongest single behavioral pattern in the dataset, directly reflecting the "women and children first" evacuation protocol.*

## Question 3: Which passenger class had the highest survival rate?

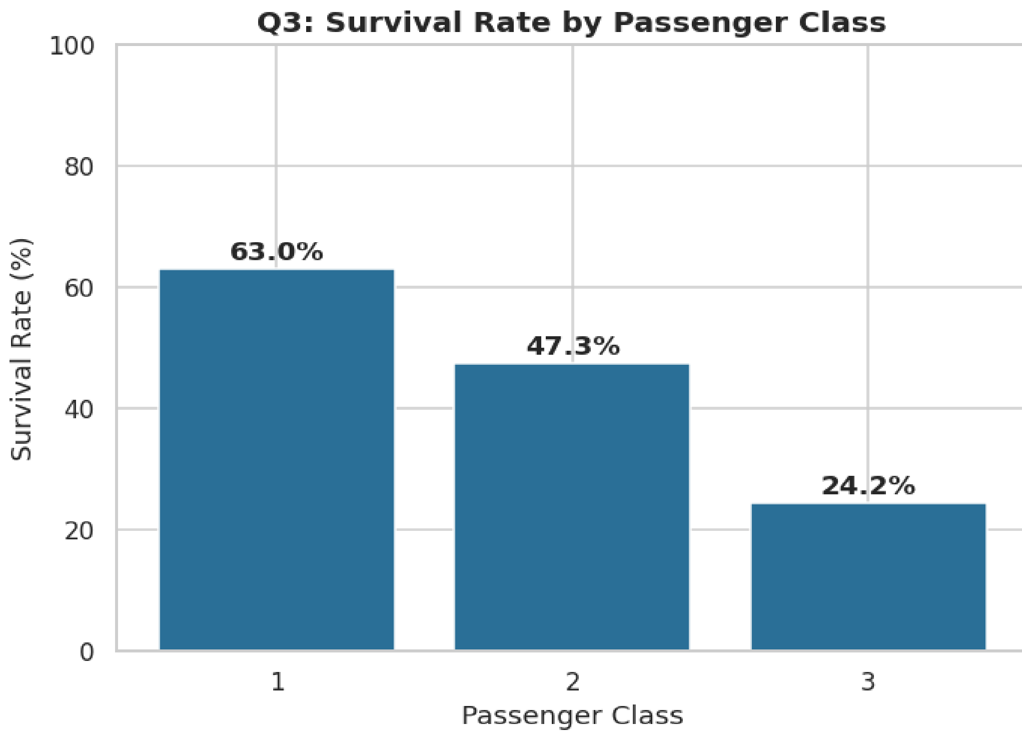


*Figure 3. Survival rate by passenger class.*

***Observation:*** *1st class passengers survived at 63.0%, compared to 47.3% for 2nd class and just 24.2% for 3rd class. Cabin location and proximity to the lifeboat deck gave wealthier passengers a clear survival advantage.*

## Question 4: Does age affect survival?

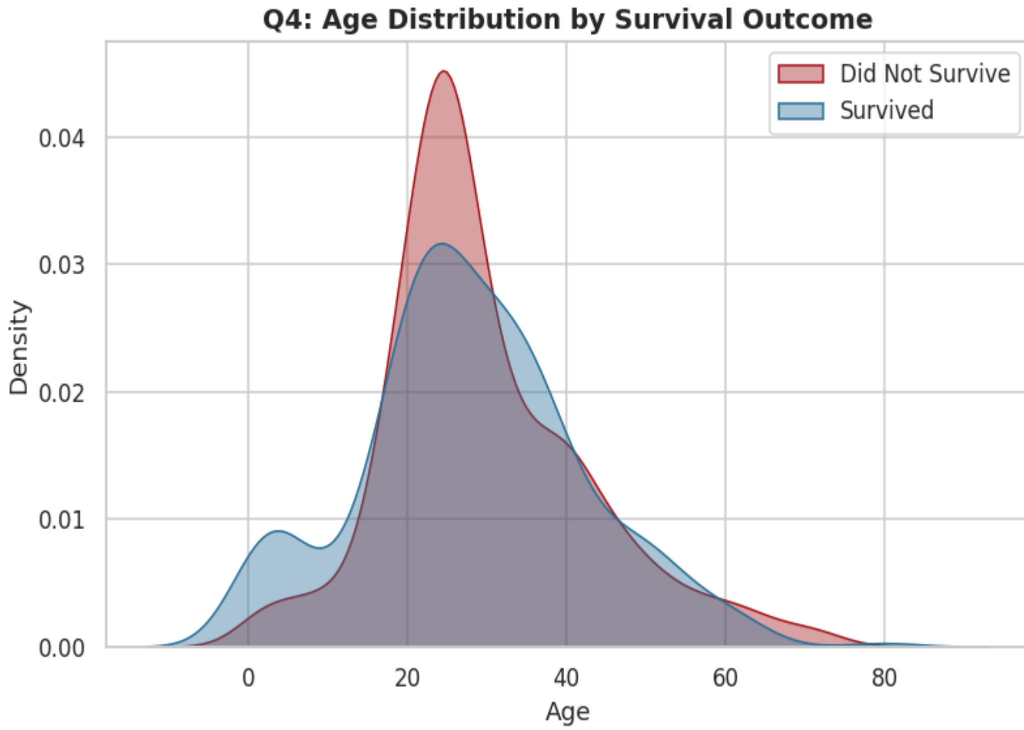


*Figure 4. Age distribution by survival outcome.*

***Observation:*** *Children had the highest survival rate of any age group (58.0%), while senior citizens had the lowest (26.9%). Age clearly mattered, reinforcing the same "children first" priority seen in the gender analysis.*

## Question 5: Does fare influence survival?

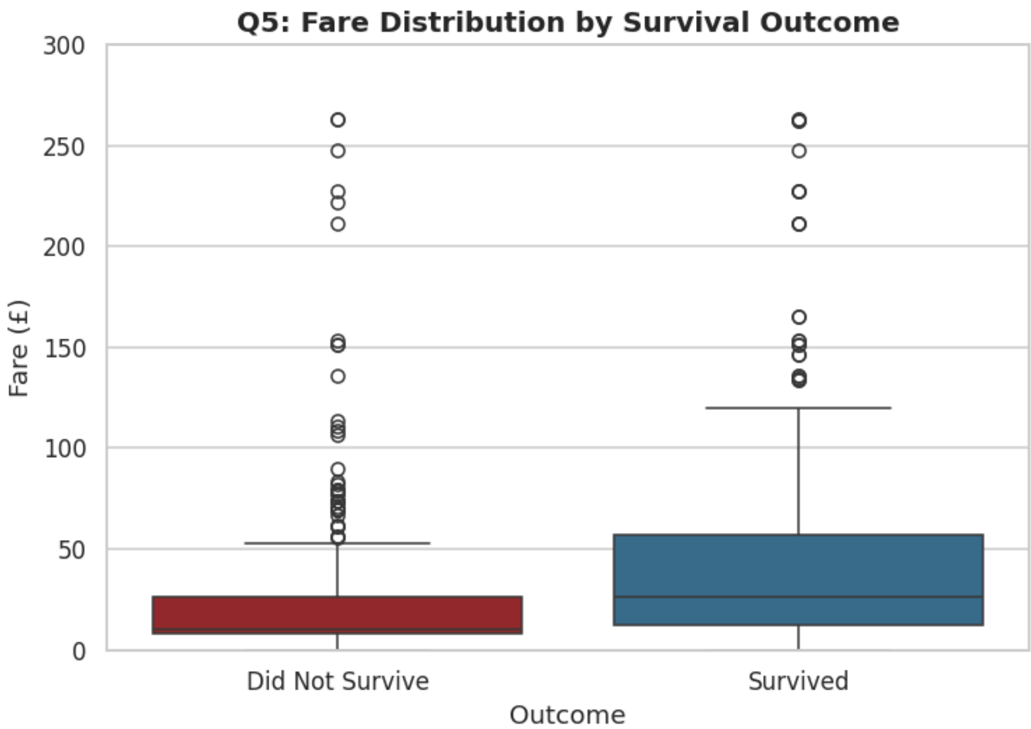


*Figure 5. Fare distribution by survival outcome.*

***Observation:*** *Survival rises sharply with fare band: 24.9% for Low Fare, 46.9% for Medium Fare, and 67.2% for High Fare passengers. Since fare tracks passenger class, this confirms wealth was strongly linked to survival chances.*

## Question 6: Did passengers travelling alone survive more often?

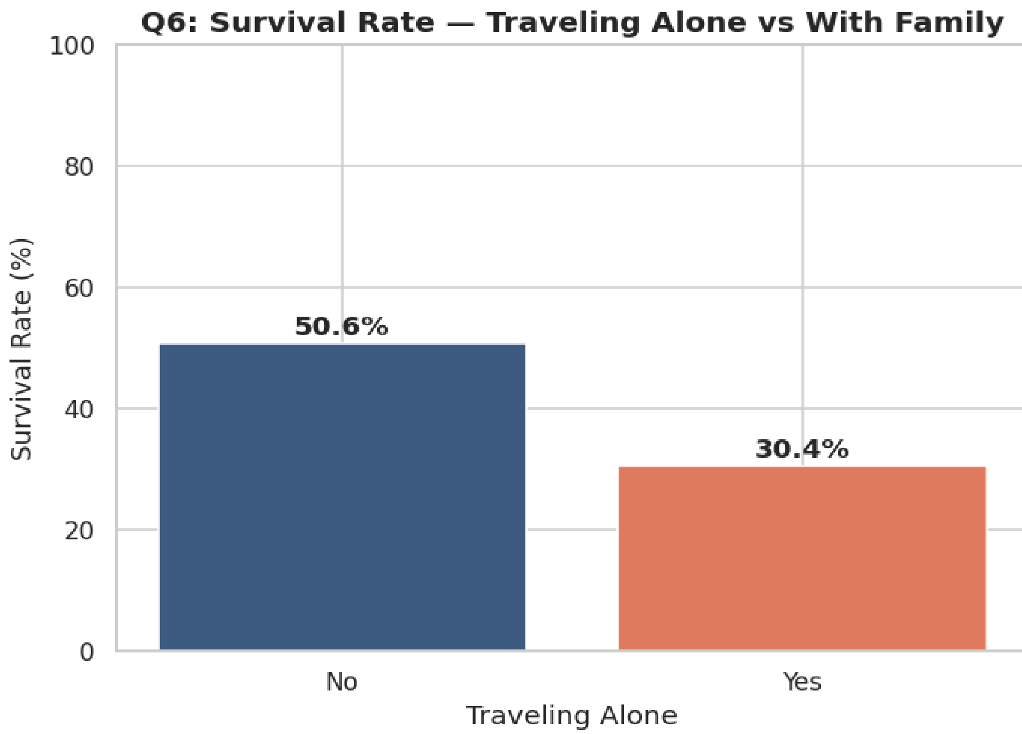


*Figure 6. Survival rate — traveling alone vs. with family.*

***Observation:*** *Passengers traveling with family survived at 50.6%, notably higher than the 30.4% survival rate of those traveling alone. Small family groups likely helped each other reach lifeboats faster.*

## Question 7: Which embarkation port had the highest survival rate?

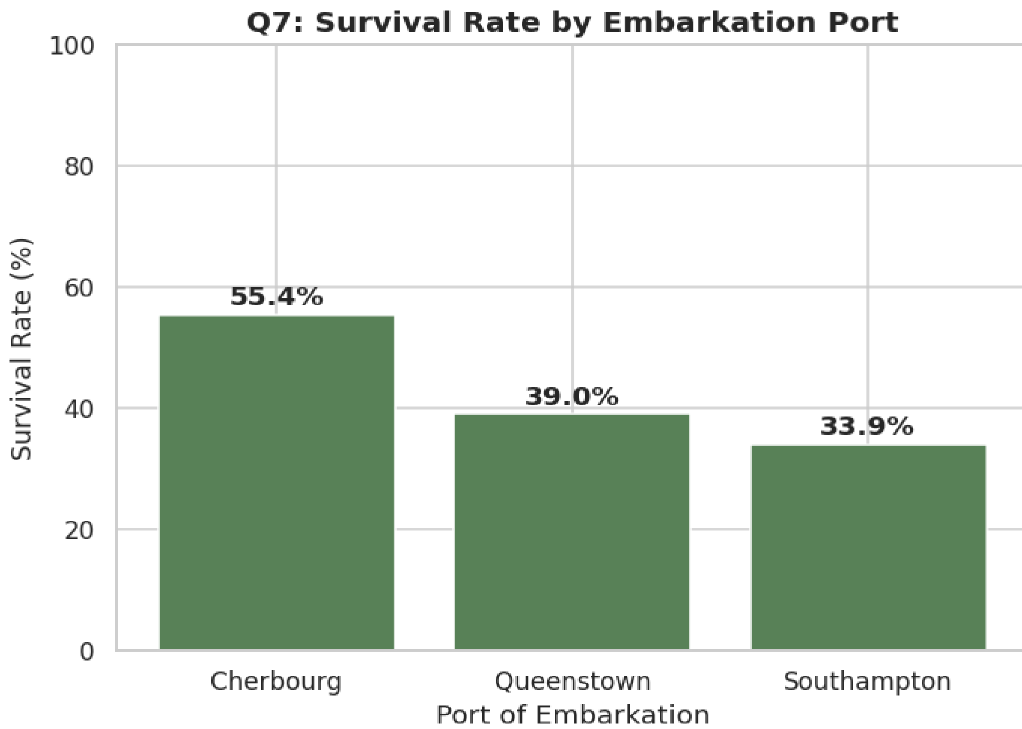


*Figure 7. Survival rate by port of embarkation.*

***Observation:*** *Passengers who boarded at Cherbourg survived at the highest rate (55.4%), compared to Queenstown (39.0%) and Southampton (33.9%) — largely because Cherbourg had a higher proportion of 1st class passengers.*

## Question 8: What is the relationship between passenger class and fare?

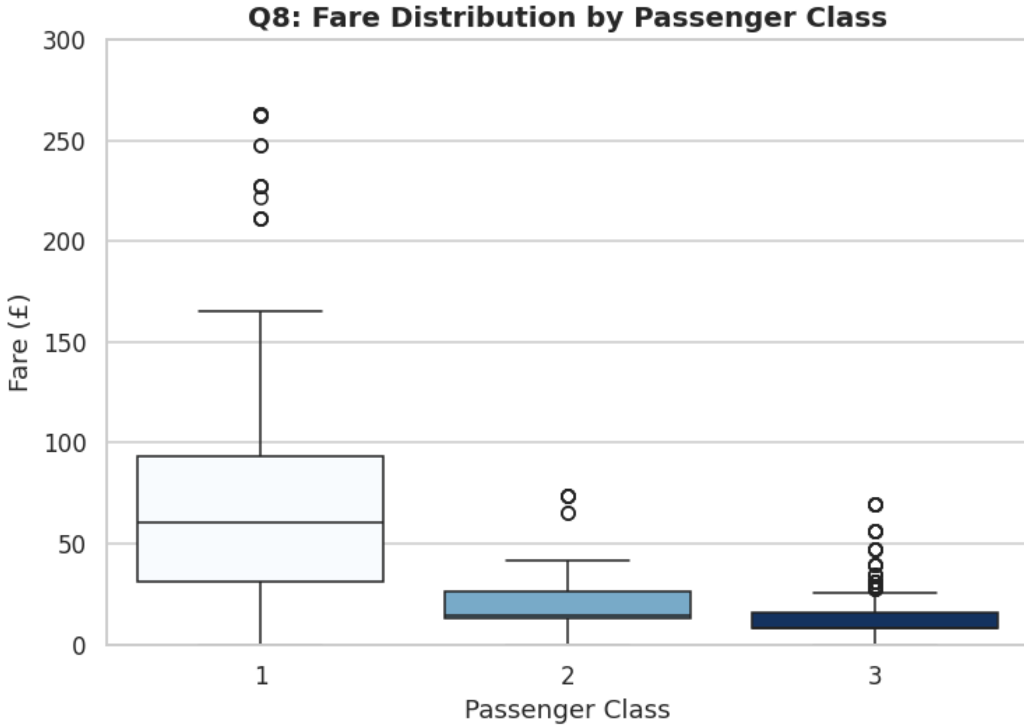


*Figure 8. Fare distribution by passenger class.*

***Observation:*** *Average fare falls sharply as class number increases: £84.15 for 1st class, £20.66 for 2nd class, and £13.68 for 3rd class. Fare is essentially a direct proxy for passenger class.*

## Question 9: Correlation Heatmap

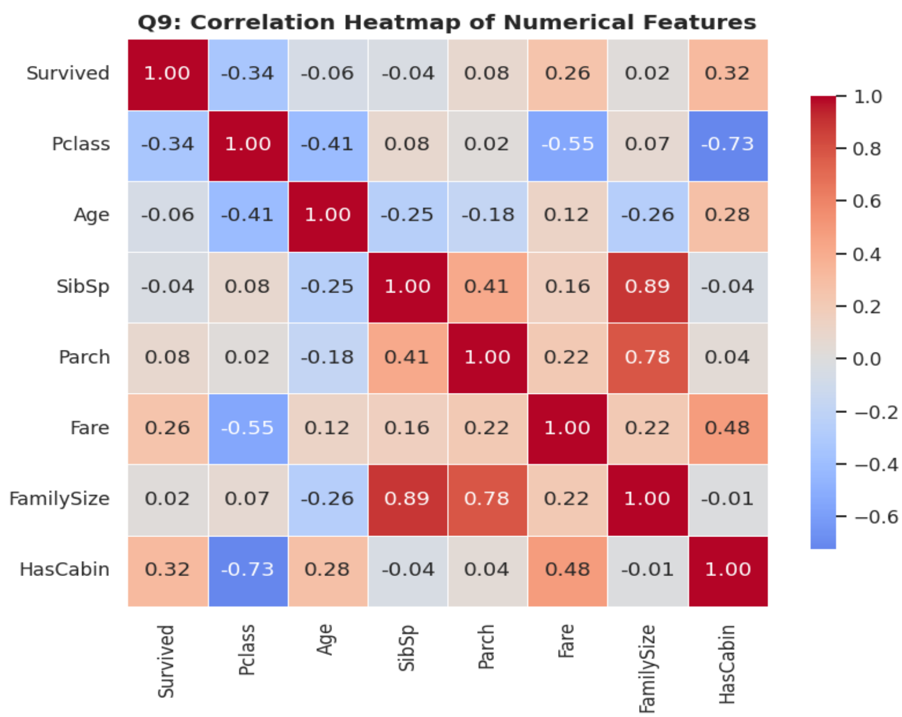

*Figure 9. Correlation heatmap of numerical features.*

***Observation:*** *The strongest positive correlation with survival is HasCabin (r = 0.32), followed by Fare (r = 0.26). The strongest negative correlation is Pclass (r = -0.34), confirming that a higher class number meant a lower chance of survival. Age shows only a weak linear correlation (-0.06) on its own — its effect is clearer once segmented into age groups (see Question 4).*

## Question 10: Custom Analysis — Family Size vs. Survival

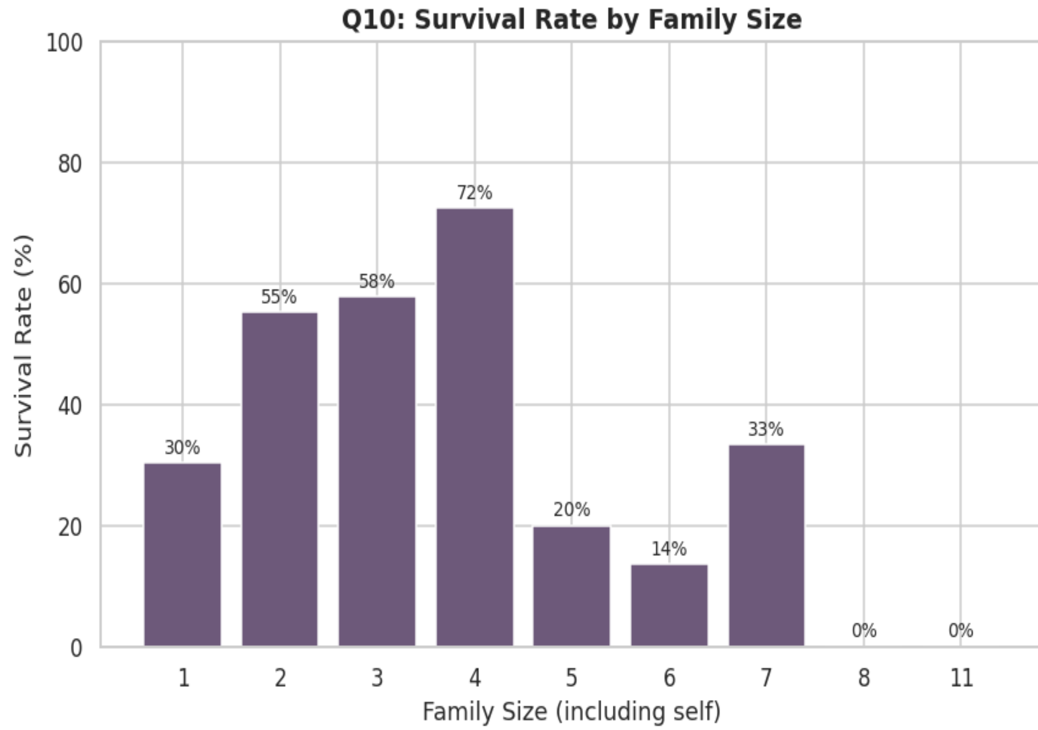


*Figure 10. Survival rate by family size.*

***Observation:*** *Survival follows an inverted-U shape across family size. Solo travelers survived at just 30.4%, but survival climbed to 55.3%–72.4% for families of 2–4, before collapsing again for families of 5 or more (as low as 0% for families of 8 or 11). Mid-sized families likely helped each other during evacuation, while very large families struggled to stay together and board in time.*

# Part 4: Statistical Analysis

Descriptive statistics were computed for three key numerical columns: Age, Fare, and FamilySize.

| Column | Mean | Median | Mode | Std. Dev. |
|---|---|---|---|---|
| Age | 29.11 | 26.00 | 25.0 | 13.30 |
| Fare | 32.20 | 14.45 | 8.05 | 49.69 |
| FamilySize | 1.90 | 1.00 | 1 | 1.61 |

## Age

Mean 29.11, Median 26.00, Mode 25.0, Std Dev 13.30. The mean sits above the median, showing a mild right skew — a smaller number of older passengers pull the average up. A standard deviation of ~13 years reflects a fairly wide age spread aboard.

## Fare

Mean 32.20, Median 14.45, Mode 8.05, Std Dev 49.69. The large gap between mean and median, and a standard deviation larger than the mean itself, confirms fare is heavily right-skewed: most passengers paid modest fares, while a small number of expensive 1st class tickets pull the average far above the typical value.

## Family Size

Mean 1.90, Median 1.00, Mode 1, Std Dev 1.61. Most passengers traveled alone (median and mode both equal 1), but the mean above 1 and a std dev above 1.6 shows a meaningful minority traveled in larger family groups, some quite large.

# Part 5: Dashboard

The dashboard below combines the six most important visualizations into a single executive-level summary view of the dataset.

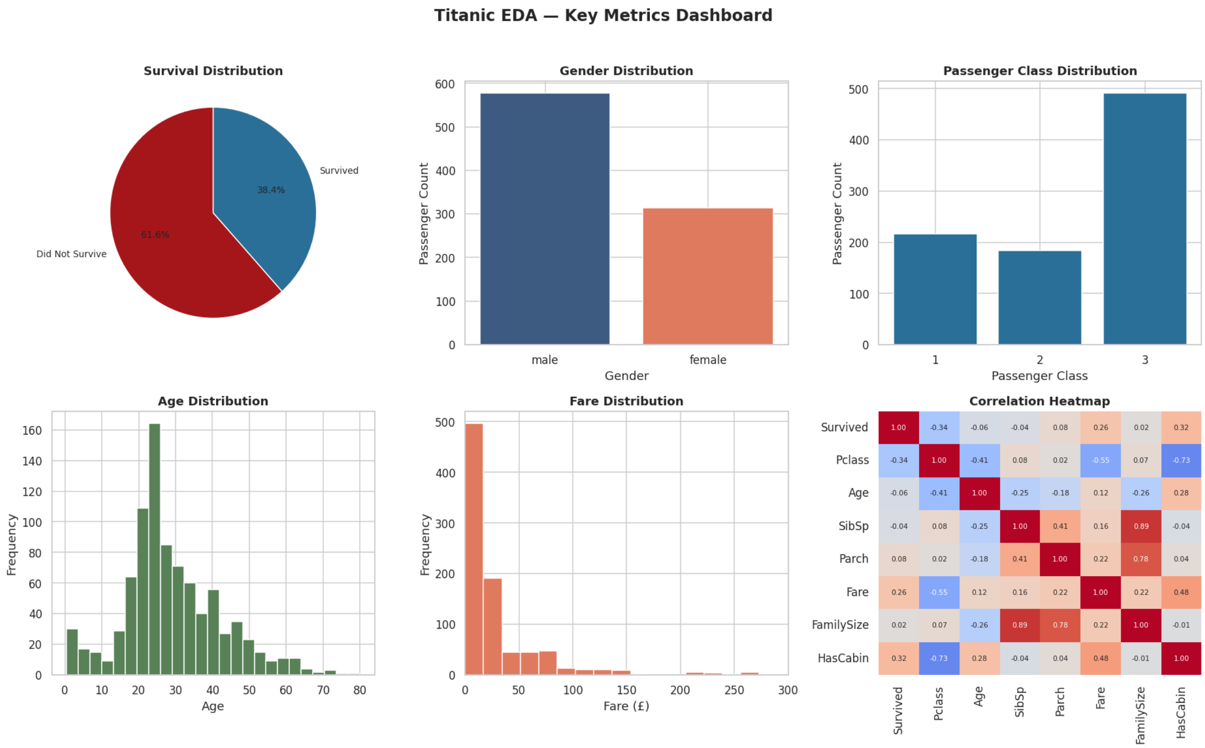


*Figure 11. Titanic EDA key-metrics dashboard.*

# Part 6: Business Insights

1. Overall survival was low — only 38.4% of all passengers survived, meaning the disaster was fatal for the large majority aboard.

2. Gender was the single biggest survival factor. Women survived at 74.2% versus 18.9% for men, a nearly 4x gap, reflecting strict "women and children first" boarding priority.

3. Passenger class strongly predicted survival. 1st class passengers survived at 63.0%, more than double the 3rd class rate of 24.2%.

4. Children were prioritized and it shows in the data. Children had the highest survival rate of any age group (58.0%), while senior citizens had the lowest (26.9%).

5. Fare tracked survival almost as closely as class. High Fare passengers survived at 67.2% versus 24.9% for Low Fare passengers — money bought both comfort and a better chance of survival.

6. Small families fared better than solo travelers or very large families. Survival peaked at family sizes of 2–4 (55–72%) and collapsed for families of 5 or more, some groups recording 0% survival.

7. Port of embarkation is a proxy for class, not a cause on its own. Cherbourg's higher survival rate (55.4%) lines up with it having a larger share of 1st class passengers than Southampton or Queenstown.

8. Having a recorded cabin was the strongest numeric correlate of survival (r = 0.32), even ahead of fare — likely because a cabin location put passengers physically closer to the lifeboat deck.

9. 3rd class men were the most vulnerable subgroup in the entire dataset, surviving at just 13.5%, far below every other class/gender combination.

10. Women survived at over 90% in 1st and 2nd class (96.8% and 92.1% respectively), only dropping to 50.0% in 3rd class — showing that even the strong gender effect was still moderated by class.

# Part 7: Business Recommendations

Presented as if to the shipping company's management, based on the evidence above.

1. Redesign evacuation drills around demonstrated at-risk groups. Since 3rd class male passengers survived at only 13.5%, evacuation planning and crew training should specifically ensure lower-deck and lower-fare passengers get equal, timely access to lifeboats and muster stations — not just those in higher-class cabins.

2. Improve wayfinding and crew presence in lower-class sections. The strong link between cabin location (HasCabin) and survival (r = 0.32) suggests physical distance from lifeboats was a major bottleneck; clearer signage and dedicated crew routing for 2nd/3rd class decks would reduce that gap.

3. Build family-assistance protocols for both very small and very large groups. Solo travelers (30.4% survival) and large families of 5+ (as low as 0%) were the two worst-performing groups by family size — staff should be trained to actively assist solo passengers and help large families stay together and board as a unit rather than being split up.

4. Do not rely on embarkation port as an operational signal — treat it as a symptom of class mix. Since the port effect is really a class effect, resourcing decisions should be driven by passenger class and deck location, not by which port passengers boarded at.

5. Track and report survival-relevant risk factors (class, deck, family size) as standard safety KPIs. Formalizing metrics like "% of 3rd class passengers with lifeboat access within X minutes" would let the company measure whether future safety investments are closing the gaps identified in this analysis.

## Conclusion

This analysis confirms that survival on the Titanic was not random — it was driven by a combination of gender, class, wealth (fare/cabin), age, and family size. Women, children, 1st class passengers, and those with a recorded cabin consistently had the highest survival rates, while 3rd class men traveling alone had by far the lowest. These patterns form the evidence base for the recommendations above and demonstrate how exploratory data analysis can translate historical data into actionable business guidance.